SYSTEMATIC LITERATURE REVIEW · MIDTERM PROJECT REPORT

Smart Traffic Lights That Learn

A Systematic Literature Review of Reinforcement Learning for Traffic Signal Control: From Simulation to Real Streets

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19 October 2026
Status: design phase complete — approximately 50 % of the project

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Summary

Project. *Smart Traffic Lights That Learn* is a systematic literature review (SLR) of reinforcement learning (RL) for traffic signal control. RL is the trial-and-error form of artificial intelligence in which a traffic light learns its own timing from a reward, such as "less waiting". Hundreds of studies report that such controllers reduce waiting times, but almost all of them were tested in computer simulations. Our review asks how close self-learning traffic lights really are to use on real streets, and what stands in the way.

Method. We follow PRISMA 2020, the international standard for conducting and reporting systematic reviews. The review covers peer-reviewed journal and conference papers published from 2019 to 2026 and indexed in Scopus, Web of Science and IEEE Xplore. Two reviewers screen every paper independently, rate its quality on a 10-point scale, and extract data into a fixed codebook.

Main idea. Besides the usual questions (which methods, how much improvement), we rate every study on a six-level **Validation Realism Level (VRL)** scale. It runs from a purely synthetic simulation (VRL 0) to a sustained deployment in a city (VRL 5). This lets us test whether the reported gains shrink as the tests become more realistic.

What has been done.

- The topic, scope, objectives and six research questions are defined, and the full review protocol is written.
- A scoping search identified the existing reviews, the vocabulary of the field, the dominant simulators and the first visible gaps.
- Search strings for all three databases, a 20-paper pilot set, the eligibility criteria, the quality rubric and a 64-field extraction codebook are ready.
- A data-management toolkit for screening, PRISMA counting and analysis has been built and tested.

Status. The project is about **50 % complete**. The design half of the review is finished. The execution half (running the database searches, screening, data extraction, synthesis and writing) starts next. Section 9 explains how the percentage was calculated.

1. Introduction

1.1 The problem

Most traffic lights in cities still run on fixed timers: green and red phases last the same time whether the road is empty or jammed. *Actuated* signals react to detectors, but they still follow rules written by engineers. A learning traffic light instead observes real traffic through cameras and road sensors and adapts its timing to what it has learned works best.

Reinforcement learning is one of the most popular ways to build such a controller. The controller (the *agent*) tries different decisions, receives a *reward* when vehicles wait less, and gradually learns a better *policy*. Research on RL traffic signal control has grown quickly since 2019, and many papers report shorter waiting times, queues and travel times. However, nearly all of these results come from simulation, and peer-reviewed tests at real intersections remain very rare. Understanding why this gap exists is the purpose of our review.

1.2 Why this is a smart system

A self-learning traffic light is a complete smart system that works in a continuous loop:

Step	What happens
Sense	Cameras, loop detectors or radar measure queues and waiting vehicles
Decide	The RL agent, usually a neural network, chooses the next green phase or its duration
Act	The signal switches at the intersection
Learn	A reward (for example, less waiting) is used to improve the policy

The same loop explains the difficulty. Trial and error is cheap and safe in a simulator, but at a real intersection a poor decision delays real people and can create safety risks.

1.3 Why a systematic review

A systematic review collects, filters and analyses all relevant published studies using a fixed, reproducible procedure. Unlike a narrative review, every step (search strings, inclusion rules, quality scoring, data extraction) is defined in advance and reported, so another team could repeat it and obtain the same evidence base.

2. Feasibility and positioning

2.1 Feasibility assessment

Question	Assessment
Is there enough published research?	Yes. The field is very active (KDD, CIKM, AAAI, NeurIPS, IEEE T-ITS, IEEE ITSC, Transportation Research Part C, Expert Systems with Applications). One earlier review alone found 160 papers on network-level RL signal control up to March 2020.
Is the topic important?	Yes. Congestion costs time, fuel, emissions and safety, and signal control is a low-cost lever because the infrastructure already exists.
Has a similar recent SLR been published?	Partly. The closest systematic review (Noaeen et al., 2022) ends in March 2020. Later reviews are narrative or focus on other angles (see 2.2). None codes how realistically each study was validated.
Can we add something new?	Yes. We cover 2019–2026, introduce the Validation Realism Level scale, relate reported gains to baseline strength and validation realism, and control for papers that reuse the same data.
Can the topic be searched systematically?	Yes. Two well-defined concepts (RL and traffic signals) with stable vocabulary.
Is quantitative comparison possible?	Yes, descriptively. Most papers report travel time, waiting time, delay or queue length against standard baselines. A formal meta-analysis is unlikely to be possible (see 8.4).

2.2 Existing reviews found in the scoping search

Review	Type	Coverage	How our review differs
Noaeen et al. (2022), *Expert Systems with Applications*	Systematic review	160 papers, 1994 – March 2020, network-level control	We cover 2019–2026 and code validation realism
Haydari & Yilmaz (2022), *IEEE Trans. on Intelligent Transportation Systems*	Narrative survey	Deep RL across intelligent transportation	Background for our taxonomy
Wei, Zheng, Gayah & Li (2019), *A Survey on Traffic Signal Control Methods*	Narrative survey	Classical and RL methods	Background for method families
Zhang, Cui & Ma (2024), *Transport Reviews*	Systematic review	RL signal control with connected and automated vehicles	Different focus
Michailidis et al. (2025), *Infrastructures*	Review	High-impact RL signal-control methods	Not focused on deployment
Chen, Fang & Sadeh (2022), *"The Real Deal"*	Position paper	Barriers to real-world deployment	Motivates our barrier question; not systematic

3. Preliminary scoping search: what we learned

Before designing the formal search, we ran an exploratory scoping search (17 October 2026, nine queries in a general web search engine). Its purpose was to learn the field's vocabulary, find existing reviews and spot likely gaps. **These findings shaped our design. They are not review results and are not counted in the PRISMA numbers.**

3.1 Vocabulary of the field

Concept	Terms used in the literature
Reinforcement learning	reinforcement learning, deep RL, multi-agent RL (MARL), Q-learning, deep Q-network (DQN, DDQN, D3QN), actor–critic (A2C, A3C), policy gradient, PPO, DDPG, SAC, meta-RL, hierarchical RL, offline RL
Traffic signals	traffic signal control (TSC), adaptive traffic signal control (ATSC), traffic light control, signalised/signalized intersection, signal timing, signal phase, cycle length, green split, arterial coordination
Deployment	real-world deployment, field test or trial, sim-to-real, transfer, grounded action transformation, domain randomisation, hardware-in-the-loop, digital twin
Emerging topics	large language model (LLM) agents for signal control, safe RL and action masking, robustness to detector failure

3.2 Observations

1. **Simulators dominate.** Recent reviews name SUMO and CityFlow as the main test platforms. Other platforms are VISSIM, Aimsun and the benchmark suites RESCO and LibSignal.
2. **Field tests are rare.** Deployment-focused papers state that RL signal controllers have essentially not yet been deployed on live streets (Chen, Fang & Sadeh, 2022; Müller & Sabatelli, 2022), and that most work uses simplified simulation environments (Müller et al., 2021). Searches aimed specifically at field trials found mainly simulations that use real traffic data.
3. **The last systematic review stops in 2020.** Recent years, including the sim-to-real methods (2023–2024) and LLM-based controllers (2024–2025), have not been reviewed systematically.
4. **The same data is reused many times.** City datasets such as Jinan (3×4 intersections), Hangzhou (4×4) and New York, and the Monaco, Ingolstadt and Cologne scenarios, appear in many papers. Studies are therefore not independent, and our analysis must avoid counting the same evidence twice.
5. **Typical design and outcomes.** A typical paper proposes an algorithm and compares it with baselines in simulation. Baselines are fixed-time plans, actuated control, Max-Pressure, or other RL methods. Reported outcomes include travel time, waiting time, delay, queue length, throughput, stops, emissions and safety measures. Variation between repeated runs is reported inconsistently.

3.3 Landmark method families identified

Family	Representative work
Single-intersection deep Q-learning	Early DQN controllers
Phase competition	FRAP (2019)
Pressure-based rewards and states	PressLight (2019), MPLight (2020), "Expression might be enough" (2022)
Graph and attention coordination	CoLight (2019), CoSLight (2024)

Family	Representative work
Multi-agent actor–critic	MA2C (2020)
Meta-learning and universal models	MetaLight (2020), AttendLight (2020)
Sim-to-real transfer	UGAT (2023), Sim2Real for TSC (2023), PromptGAT (2024)
LLM-based agents	LLMLight (2025)

3.4 Gaps already visible

- Field validation: very few peer-reviewed live tests.
- Robustness to detector noise or failure, and partial observation.
- Safety constraints such as minimum green time and pedestrian clearance.
- Pedestrians, cyclists, public transport and emergency vehicles.
- Interpretability and acceptance by traffic engineers.
- Comparison against strong adaptive baselines, not only fixed-time plans.
- Reproducibility: code, random seeds and scenario files are often missing.

4. Scope of the review

> *This systematic review investigates peer-reviewed primary studies concerning reinforcement-learning-based controllers applied to road traffic signals at one or more intersections, for the purpose of adaptively selecting or timing signal phases to improve traffic performance, published between 1 January 2019 and the date of the final search (2026), with particular attention to how realistically the controllers were validated, from simulation to field.*

Dimension	Included	Excluded
Subject	Controllers that *learn* their policy with RL: tabular, deep, multi-agent, hierarchical, meta, offline and model-based RL, and hybrids where RL is the decision layer (including LLM agents refined with RL)	Non-RL optimisation only (genetic algorithms, MPC, fuzzy logic), prompting-only LLM controllers, RL used only as a minor baseline
Application	Urban and suburban road intersections: isolated, corridors and networks; also mixed traffic, connected vehicles, pedestrians, cyclists, public transport and emergency priority when RL controls the signal	Ramp metering only, speed limits only, routing, autonomous driving policies, rail, air or maritime traffic, telecom "traffic signals"
Technology	Any sensing (loops, cameras, radar, connected vehicles) and any evaluation environment (SUMO, CityFlow, VISSIM, Aimsun, hardware-in-the-loop, field)	—
Geography	Worldwide, no restriction	—
Period	1 January 2019 – final search date (2026)	Earlier papers
Publication type	Peer-reviewed journal articles (including early access) and full conference papers	Reviews (used only for citation chaining), preprints, theses, book chapters, posters, extended abstracts
Language	English	Other languages (reported as a limitation)

Why 2019.

1. It gives deliberate continuity with the previous systematic review, which ends in March 2020, with a one-year overlap.
2. It marks the shift to open, large-scale tools and real-data benchmarks (CityFlow; the Jinan, Hangzhou and New York datasets) and to landmark deep-RL models (PressLight, CoLight, FRAP).
3. The deployment and sim-to-real debate appears from 2021 onwards.

5. Objectives and research questions

5.1 Main objective

To identify, classify, compare and critically evaluate peer-reviewed research (2019–2026) on reinforcement learning for traffic signal control. We pay particular attention to RL formulations, evaluation environments and validation realism, reported traffic performance, deployment barriers, and the research gaps that separate simulation results from real-street use.

5.2 Specific objectives

No.	Specific objective	Research question
SO1	Quantify publication trends, venues, countries and application contexts	RQ1
SO2	Classify RL methods: algorithm family, single vs multi-agent, state, action, reward	RQ2
SO3	Identify simulators, road networks, demand data and baselines	RQ3
SO4	Rate every study on the Validation Realism Level scale	RQ3
SO5	Compare reported improvements in waiting time, queue length and travel time or delay	RQ4
SO6	Test whether improvements depend on baseline strength and validation realism	RQ4
SO7	Catalogue advantages, limitations, deployment barriers and proposed solutions	RQ5
SO8	Assess methodological quality and reproducibility	All
SO9	Identify research gaps	RQ6
SO10	Recommend future research directions	RQ6

5.3 Research questions

The project proposal asked four questions. They were refined into six questions that can be answered from published papers.

RQ	Question	Main information extracted
RQ1 Landscape	How has research on RL traffic signal control evolved from 2019 to 2026 in volume, venues, countries and application contexts?	Year, venue, country, network size, context
RQ2 Methods	Which RL formulations are used (algorithm family, agent architecture, state, action and reward design), and how have they changed?	Algorithm family, agents, coordination, state, action, reward
RQ3 Validation	How are the methods evaluated (simulators, networks, demand data, baselines, robustness tests), and how are studies distributed across Validation Realism Levels 0–5?	Simulator, network and demand source, baselines, VRL
RQ4 Performance	What improvements in waiting time, queue length and travel time or delay are reported against fixed-time, actuated or Max-Pressure, and RL baselines, and do they differ by baseline type and validation realism?	Numerical results against each baseline
RQ5 Barriers	Which strengths, limitations and deployment barriers are reported, and which solutions are proposed? Barriers include sensing, the sim-to-real gap, safety, robustness, computation, communication, interpretability, multimodality and scalability.	Barrier categories and mitigations

RQ	Question	Main information extracted
RQ6 Gaps	Which gaps remain, and which approaches look most promising for moving from simulation to real streets?	Gap matrix and future-work statements

5.4 Validation Realism Level (VRL): the central rating scale

Level	Name	Definition (highest level the study reaches)
VRL 0	Synthetic simulation	Synthetic network (for example, a grid) and synthetic traffic
VRL 1	Real-geometry simulation	Real road layout, but synthetic or uncalibrated traffic
VRL 2	Real-data simulation	Simulation driven by observed traffic data or calibrated against field measurements
VRL 3	Hardware-in-the-loop or shadow mode	Real controller hardware or live sensor feeds in the loop, but the RL policy does not switch street signals
VRL 4	Field test	The RL policy switches real signals at one or more real intersections during a limited trial
VRL 5	Sustained deployment	The RL controller has run continuously for at least one month, with reported outcomes

Each study also receives three yes/no flags: *transfer test* (tested on unseen scenarios or another simulator), *robustness test* (detector noise or failure, demand surges, incidents) and *safety constraints* (minimum green time, clearance times, action masking).

6. Review protocol

6.1 Framework and team

The review follows **PRISMA 2020** (27-item checklist, abstract checklist and flow diagram) for reporting, and **PRISMA-S** for reporting the searches. The protocol (version 1.0, October 2026) will be registered on the Open Science Framework before the final searches are run. Any later change will be recorded as a dated amendment with its reason.

Task	Reviewer 1: Rakhimzhanov Amangeldy	Reviewer 2: Nurkin Aidar
Running the searches	Scopus, IEEE Xplore	Web of Science, citation chaining
Search log	Owner	Checks
Title and abstract screening	Calibration sample + 50 %	Calibration sample + 50 %
Full-text screening	100 %	100 %
Quality assessment	100 %	100 %
Technical data extraction	Owner	Checks 20 %
Numerical data extraction	Owner	Checks 100 %
Synthesis and writing	Shared	Shared

Disagreements are resolved by discussion. If they remain unresolved, they are decided with the supervisor.

6.2 Inclusion criteria (all must be met)

Code	Criterion
IC1	Primary study that proposes, adapts or evaluates an RL-based controller for road traffic signals (phase choice, phase duration, cycle, split or offset) at one or more intersections
IC2	Reports an empirical evaluation (simulation, hardware-in-the-loop or field) with at least one numerical traffic performance outcome
IC3	Published from 1 January 2019 to the final search date
IC4	Peer-reviewed journal article or full conference paper
IC5	Written in English
IC6	Full text obtainable by the team

6.3 Exclusion reasons (one primary reason per excluded paper)

Code	Reason
E1	Wrong application: not road traffic signal control
E2	Wrong method: RL is not the evaluated controller
E3	Wrong publication type: review, survey, position paper, book chapter
E4	No empirical evaluation or no numerical traffic outcome
E5	Outside the date range
E6	Not peer-reviewed: preprint, thesis, poster, extended abstract

Code	Reason
E7	Not in English
E8	Duplicate record missed at deduplication
E9	Full text not obtainable after documented attempts
E10	Not enough technical information to identify the RL design or results

When several reasons apply, the first applicable code in this order is recorded.

6.4 Selection procedure

1. **Deduplication.** Records from the three databases are merged. Records are combined automatically only when the DOI is the same, or the title, year and first author are identical. Similar-looking titles are checked by hand. A conference paper and its extended journal version are kept as separate reports but linked as one study (see 8.5).
2. **Title and abstract screening.**
   - **Calibration.** Both reviewers first screen the same random 10 % of records (at least 50) and compute Cohen's kappa. The target is 0.60 or higher. If it is lower, the criteria are discussed, clarified and re-tested.
   - **Remaining records.** These are split between the reviewers. Any exclusion a reviewer is unsure about is checked by the other.
   - **Decisions.** Each record is marked *include*, *exclude* or *uncertain*. If either reviewer says *include* or *uncertain*, the paper goes to full-text screening.
3. **Full-text screening.** Both reviewers assess every full text independently. Each excluded paper receives one exclusion code. Disagreements are settled by discussion and the outcome is recorded.
4. **PRISMA flow diagram.** All counts in the diagram are computed from the screening records, never estimated: records identified per database, duplicates removed, records screened and excluded, reports not retrieved, exclusions by reason, and studies and reports included.

7. Search strategy

7.1 Concept matrix

Concept	Role	Terms
A: Reinforcement learning	Method	reinforcement learning, Q-learning, deep Q-network, DQN, actor-critic, policy gradient, proximal policy optimization
B: Traffic signals	Application	traffic signal(s), traffic light(s), signalized/signalised intersection(s), signal timing, signal phase, adaptive signal control
C: Deployment	Targeted searches only	field test or trial, real-world deployment, sim-to-real, hardware-in-the-loop, digital twin, real intersection, transfer learning
D: Expansion terms	Second wave	SARSA, temporal difference, DDPG, soft actor-critic, A2C/A3C, MARL, phase selection
E: Context	Gap-targeted searches	pedestrian, public transport, emergency vehicle, connected vehicle, arterial, network

The core search is **A AND B** in the title, abstract and keywords. The deployment concept (C) is deliberately *not* required in the core search: the large body of simulation-only work is exactly what RQ3 and RQ4 need to measure. "Signal control" alone was rejected because it also retrieves power-electronics and telecommunications papers, and "intersection" alone was rejected because it is too broad.

7.2 Databases

Database	Why
Scopus	Broadest multidisciplinary coverage, including ACM (KDD, CIKM), AAAI, IEEE, Elsevier, Springer, MDPI and Taylor & Francis
Web of Science Core Collection	Independent curated index, strong for transport journals
IEEE Xplore	Subject-specific: IEEE T-ITS, ITSC, CDC and CASE, including early-access papers

Google Scholar is used only for forward citation chaining and is not counted as a database. arXiv is not searched because preprints are excluded; it is used only to locate the published version of a paper.

7.3 Exact search strings

S1 — Scopus (searched fields: TITLE-ABS-KEY)

```
TITLE-ABS-KEY ( ( "reinforcement learning" OR "Q-learning" OR "deep Q-network*" OR "DQN" OR "actor-critic" OR "policy gradient" OR "proximal policy optimi*" ) AND ( "traffic signal*" OR "traffic light*" OR "signali?ed intersection*" OR "signal timing" OR "signal phas*" OR "adaptive signal control" ) ) AND PUBYEAR > 2018 AND PUBYEAR < 2027 AND ( LIMIT-TO ( DOCTYPE , "ar" ) OR LIMIT-TO ( DOCTYPE , "cp" ) ) AND ( LIMIT-TO ( LANGUAGE , "English" ) )
```

Filters: PUBYEAR 2019-2026 (in query); DOCTYPE ar, cp; LANGUAGE English

S2 — Web of Science Core Collection (searched fields: TS)

```
TS=( ("reinforcement learning" OR "Q-learning" OR "deep Q-network*" OR DQN OR "actor-critic" OR "policy gradient" OR "proximal policy optimi*") AND ("traffic signal*" OR "traffic light*" OR "signali?ed intersection*" OR "signal timing" OR "signal phas*" OR "adaptive signal control") ) AND PY=(2019-2026)
```

Filters: Editions SCI-EXPANDED, SSCI, ESCI, CPCI-S; DT Article, Proceedings Paper, Early Access; LA English

S3 — IEEE Xplore (searched fields: All Metadata)

```
("All Metadata":"reinforcement learning" OR "All Metadata":"Q-learning" OR "All Metadata":"deep
Q-network" OR "All Metadata":"DQN" OR "All Metadata":"actor-critic" OR "All Metadata":"policy
gradient" OR "All Metadata":"proximal policy optimization") AND ("All Metadata":"traffic signal"
OR "All Metadata":"traffic signals" OR "All Metadata":"traffic light" OR "All Metadata":"traffic
lights" OR "All Metadata":"signalized intersection" OR "All Metadata":"signalized intersections"
OR "All Metadata":"signalised intersection" OR "All Metadata":"signal timing" OR "All
Metadata":"signal phase" OR "All Metadata":"adaptive signal control")
```

Filters: Year 2019-2026; Conferences, Journals, Early Access Articles

7.4 Pilot test

Before the search is locked, it is tested against a benchmark set of 20 papers that any adequate search should find (Appendix A). The set mixes landmark methods, deployment and sim-to-real papers, benchmarks, an LLM-based controller and three reviews.

Pass criterion. At least 90 % of the benchmark papers that are indexed in at least one database must be retrieved by the three searches combined. A random sample of 100 retrieved titles is also checked to estimate precision.

If a paper is missed. The cause is recorded. If a search term was missing, it is added and the pilot is repeated. The search is never tuned only to find preferred papers.

7.5 Multi-wave search plan

Wave	Purpose	How
1	Core database search	Run the locked strings in all three databases
2	Terminology expansion	Add the expansion terms (D) and deployment terms (C) found during wave 1
3	Citation chaining	Backward (reference lists) and forward ("cited by") searching from high-quality included studies and from the existing reviews
4	Gap-targeted and update search	Searches for under-represented contexts (pedestrians, public transport, field tests), then an update run just before submission

Every wave is logged with its date, exact string, filters and number of records, as PRISMA-S requires.

8. Assessment, extraction and analysis plan

8.1 Quality assessment

Each included study is scored on five criteria, 0 = no, 1 = partly, 2 = yes (maximum 10). Both reviewers score every study and agreement is reported.

Criterion	2 (yes)	1 (partly)	0 (no)
QA1 Research design	Objective and full RL formulation (state, action, reward, agents) clearly described	One element vague	Unclear
QA2 Data quality	Real road network and real or calibrated traffic, fully described	Synthetic but fully described, or real but poorly described	Not described
QA3 Reproducibility	Training details, simulator and version reported, and code or scenario files available	Details given but no code	Not enough to repeat
QA4 Validation	At least one adaptive non-RL baseline, plus a strong RL baseline, an unseen-scenario test or a field test	Only fixed-time or one baseline	No baseline
QA5 Reporting	Several runs with spread (SD or CI) and limitations discussed	One of the two	Neither

The thresholds are fixed in advance: **High 8–10, Moderate 5–7, Limited 0–4**. Quality is not used to exclude studies. It guides interpretation and a sensitivity analysis that leaves out Limited studies.

8.2 Data extraction

A codebook of **64 fields** describes every study in the same way. It has 45 descriptive fields per paper and 19 fields for every reported number; the full codebook is in Appendix B. Values are chosen from fixed category lists, so studies can be counted and compared.

Rules for numbers:

- One row per value pair (RL result and baseline result).
- Every number records the page, table or figure it came from.
- Results from different experiments are never averaged by the reviewers.
- Reviewer 2 checks every numerical entry.

8.3 Synthesis

- **Taxonomy** of the field: application context, road users, RL family, agents, coordination, state, action, reward, simulator, baselines, validation realism, barriers.
- **Comparison of method families.** Table 8.6 is the starting version, to be updated with evidence.
- **Descriptive analysis:** papers per year, venues, countries, simulators, VRL distribution, a heat map of RL family against VRL, and barrier frequencies.
- **Quantitative synthesis.** The relative improvement is calculated as **RI = (baseline − RL) / baseline × 100 %** for "lower is better" measures such as travel time, and **(RL − baseline) / baseline × 100 %** for throughput. A positive RI always means RL is better. Results are summarised (median, interquartile range, minimum, maximum) by measure, baseline type and validation level. Different measures are never pooled together.

8.4 Is a meta-analysis appropriate?

A meta-analysis is only run for a combination of outcome and baseline type where **at least five independent studies** report the mean, standard deviation and number of runs in the same unit. In that case we use a random-effects model on the log ratio of means and report heterogeneity (τ^2, I^2). From the scoping search we expect most combinations not to meet this rule. The review will then use a structured narrative synthesis with subgroup comparison. A meta-analysis will not be forced.

8.5 Avoiding double counting

- Papers reporting the same experiments or project, such as a conference paper and its journal extension, share a **study family ID** and count as one study.
- For the statistics, each study family contributes one value: the median of its results.
- Reuse of benchmark datasets is counted and reported.
- Families contributing more than 10 % of all numerical entries are checked in a sensitivity analysis.
- The PRISMA diagram reports both *studies* and *reports*.

8.6 Method families: initial comparison

RL family	Principle	Strengths	Limitations	Typical validation
Tabular RL	Look-up table of action values (Q-learning, SARSA)	Simple, interpretable, low compute	Needs coarse states; poor generalisation	Small synthetic simulations (VRL 0–1)
Value-based deep RL	Neural network estimates action values (DQN family)	Handles rich states; fits discrete phase choice	Unstable training; unsafe exploration on live roads	Real-data simulations (VRL 1–2)
Policy gradient	Directly optimises the policy (PPO, TRPO)	Stable updates (PPO); continuous durations possible	Needs many samples	Simulation (VRL 0–2)
Actor–critic	Policy and value learned together (A2C, MA2C, MAPPO)	Lower variance; suits multi-agent training	Sensitive to settings; coordination overhead	Grid and real-network simulations (VRL 1–2)
Model-based RL	Learns or uses a traffic model for planning	Sample efficient; safer exploration	Model errors; extra modelling effort	Simulation, sometimes hardware-in-the-loop (VRL 2–3)
Hierarchical RL	High-level goals, low-level signal agents	Scales to large networks	Complex design	Large-network simulations (VRL 1–2)
Meta-RL and transfer	Learns to adapt quickly to new intersections	Generalises across scenarios	Training cost	Cross-scenario simulation
Offline RL	Learns from logged data without live trial and error	No risky exploration; uses detector logs	Distribution shift	Logged data plus simulation (VRL 2–3)
LLM + RL hybrid	Language model reasons about traffic; policy refined with RL feedback	Generalisation, explanations	Latency, compute, limited safety guarantees	Real-data simulation (VRL 2)

9. Progress and completion: why the project is 50 % complete

9.1 Work completed

1. **Topic and feasibility:** topic defined, feasibility assessed, contribution identified (Sections 1–2).
2. **Scoping search:** existing reviews, vocabulary, platforms, landmark methods and first gaps documented (Section 3).
3. **Scope, objectives and research questions:** formal scope statement, 10 specific objectives, 6 research questions, Validation Realism Level scale (Sections 4–5).
4. **Review protocol (version 1.0):** eligibility criteria, exclusion codes, selection procedure, reviewer roles (Section 6).
5. **Search strategy:**
   - concept matrix, database selection and exact search strings for three databases;
   - a 20-paper pilot set and a multi-wave plan (Section 7).
6. **Assessment and analysis design:**
   - quality rubric with fixed thresholds and a 64-field extraction codebook;
   - synthesis, meta-analysis and double-counting rules;
   - a first taxonomy and method-family comparison (Section 8).
7. **Data-management toolkit:**
   - **Evidence register:** one master spreadsheet with 15 sheets, from the search log to the PRISMA counts.
   - **Processing steps:** importing database exports, removing duplicates, preparing the screening sheets for both reviewers, and calculating agreement.
   - **Outputs:** the PRISMA flow diagram, the extraction checks, and all synthesis tables and charts, generated directly from the records so that no number is typed by hand.
   - **Testing:** checked on artificial test records; all 10 automated checks pass.

9.2 Completion calculation

We divided the project into six phases and weighted them by expected effort (100 points in total). Only finished work earns points.

Phase	Weight	Status	Points earned
Planning and protocol (guideline steps 1–6)	20	Complete	20
Search strategy (steps 7–12)	15	Strings, databases and plan complete; pilot run and formal lock remaining	13
Screening, quality, extraction and synthesis procedures and tools (steps 13–27, design)	15	Complete and tested	15
Database searches and screening (steps 13–17, execution)	20	Next	0
Quality assessment and data extraction (steps 18–21, execution)	15	Not started	0
Synthesis and writing (steps 22–27, execution, manuscript)	15	First taxonomy and method comparison drafted	2
Total	100		50

Interpretation. The *design half* of the review is complete: we know exactly what to search, how to judge each paper and how to analyse the results. The *execution half* (searching, screening, extraction, synthesis and writing) starts with the pilot search.

9.3 Next steps

1. Run the pilot searches in Scopus, Web of Science and IEEE Xplore and check recall against the 20-paper benchmark set.
2. Adjust terms if necessary, lock search protocol version 1.0 and register the protocol.
3. Run the wave 1 searches, record the counts and remove duplicates.
4. Calibrate title and abstract screening (Cohen's kappa ≥ 0.60), then screen all records.
5. Full-text screening, quality assessment and data extraction.
6. Waves 2–4 (term expansion, citation chaining, update search).
7. Synthesis, PRISMA flow diagram and manuscript.

10. Limitations of the design

- Only English-language papers are included.
- Three databases are searched. Venues not indexed in them, such as some NeurIPS and ICML proceedings, may be under-represented; citation chaining partly compensates.
- Preprints and grey literature are excluded, so very recent or industrial deployments described only in reports may be missed.
- The scoping search used a general web search engine and is not reproducible. It was used only for design, never as evidence.
- Many studies reuse the same benchmark datasets, which limits how far the results apply to real cities. We report this reuse explicitly.

References (scoping search)

Bibliographic details will be checked against the publisher record before final citation.

1. Chen, Fang & Sadeh (2022). *The Real Deal: A Review of Challenges and Opportunities in Moving Reinforcement Learning-Based Traffic Signal Control Systems Towards Reality*. arXiv:2206.11996.
2. Müller et al. (2021). *Towards Real-World Deployment of Reinforcement Learning for Traffic Signal Control*. arXiv:2103.16223.
3. Müller & Sabatelli (2022). *Safe and Psychologically Pleasant Traffic Signal Control with Reinforcement Learning using Action Masking*. arXiv:2206.10122.
4. Noaeen et al. (2022). *Reinforcement learning in urban network traffic signal control: A systematic literature review*. Expert Systems with Applications.
5. Haydari & Yilmaz (2022). *Deep Reinforcement Learning for Intelligent Transportation Systems: A Survey*. IEEE Transactions on Intelligent Transportation Systems 23(1), 11–32.
6. Wei, Zheng, Gayah & Li (2019). *A Survey on Traffic Signal Control Methods*. arXiv:1904.08117.
7. Zhang, Cui & Ma (2024). *Transport Reviews* 44(6), 1187–1208.
8. Michailidis et al. (2025). *Traffic Signal Control via Reinforcement Learning: A Review on Applications and Innovations*. Infrastructures 10(5), 114. doi:10.3390/infrastructures10050114.
9. Da et al. (2023). *Uncertainty-aware Grounded Action Transformation towards Sim-to-Real Transfer for Traffic Signal Control*. arXiv:2307.12388.
10. Da et al. (2023). *Sim2Real Transfer for Traffic Signal Control*. IEEE CASE 2023.
11. Da et al. (2024). *Prompt to Transfer: Sim-to-Real Transfer for Traffic Signal Control with Prompt Learning*. AAAI 2024.
12. Mei et al. *LibSignal: An Open Library for Traffic Signal Control*. arXiv:2211.10649.
13. Ault & Sharon (2021). *RESCO: Reinforcement Learning Benchmarks for Traffic Signal Control*.
14. Wei et al. (2019). *CoLight: Learning Network-level Cooperation for Traffic Signal Control*. arXiv:1905.05717.
15. *CoSLight: Co-optimizing Collaborator Selection and Decision-making to Enhance Traffic Signal Control* (KDD 2024). arXiv:2405.17152.
16. *Multi-Agent Reinforcement Learning for Traffic Signal Control through Universal Communication Method (UniComm)*. arXiv:2204.12190.
17. Lai et al. (2025). *LLMLight: Large Language Models as Traffic Signal Control Agents*. KDD 2025. arXiv:2312.16044.
18. Egea et al. (2020). *Assessment of Reward Functions for Reinforcement Learning Traffic Signal Control under Real-World Limitations*. arXiv:2008.11634.
19. *Hierarchical reinforcement learning-based traffic signal control*. Scientific Reports (2025).
20. Page et al. (2021). *The PRISMA 2020 statement: an updated guideline for reporting systematic reviews*. BMJ 372:n71.
21. Rethlefsen et al. (2021). *PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews*. Systematic Reviews 10:39.

Appendix A. Pilot benchmark set (20 known papers)

The search must retrieve these papers before it is locked. The three reviews (G18–G20) are retrieved but later excluded (code E3); they are used for citation chaining. Bibliographic details will be verified during the pilot.

ID	Paper	First author	Year	Venue	Role in the pilot
G01	PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network	Wei	2019	KDD 2019	landmark method
G02	CoLight: Learning Network-level Cooperation for Traffic Signal Control	Wei	2019	CIKM 2019	landmark method
G03	Learning Phase Competition for Traffic Signal Control	Zheng	2019	CIKM 2019	landmark method (FRAP)
G04	Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control	Chen	2020	AAAI 2020	landmark method (MPLight)
G05	Multi-Agent Deep Reinforcement Learning for Large-Scale Traffic Signal Control	Chu	2020	IEEE Transactions on Intelligent Transportation Systems	landmark method (MA2C)
G06	MetaLight: Value-Based Meta-Reinforcement Learning for Traffic Signal Control	Zang	2020	AAAI 2020	meta-RL
G07	AttendLight: Universal Attention-Based Reinforcement Learning Model for Traffic Signal Control	Oroojlooy	2020	NeurIPS 2020	universal model
G08	Expression might be enough: representing pressure and demand for reinforcement learning based traffic signal control	Zhang	2022	ICML 2022	pressure-based state
G09	CoSLight: Co-optimizing Collaborator Selection and Decision-making to Enhance Traffic Signal Control	Ruan	2024	KDD 2024	graph/collaboration
G10	Towards Real-World Deployment of Reinforcement Learning for Traffic Signal Control	Müller	2021	ICMLA 2021	deployment
G11	Safe and Psychologically Pleasant Traffic Signal Control with Reinforcement Learning using Action Masking	Müller	2022	IEEE ITSC 2022	safety/deployment
G12	Assessment of Reward Functions for Reinforcement Learning Traffic Signal Control under Real-World Limitations	Egea	2020	IEEE SMC 2020	real-world limitations
G13	Uncertainty-aware Grounded Action Transformation towards Sim-to-Real Transfer for Traffic Signal Control	Da	2023	IEEE CDC 2023	sim-to-real

ID	Paper	First author	Year	Venue	Role in the pilot
G14	Prompt to Transfer: Sim-to-Real Transfer for Traffic Signal Control with Prompt Learning	Da	2024	AAAI 2024	sim-to-real
G15	Sim2Real Transfer for Traffic Signal Control	Da	2023	IEEE CASE 2023	sim-to-real
G16	LLMLight: Large Language Models as Traffic Signal Control Agents	Lai	2025	KDD 2025	LLM + RL hybrid
G17	LibSignal: An Open Library for Traffic Signal Control	Mei	2023	Machine Learning (journal) / NeurIPS 2022 workshop	benchmark
G18	Reinforcement learning in urban network traffic signal control: A systematic literature review	Noaeen	2022	Expert Systems with Applications	review (E3; used for chaining)
G19	Deep Reinforcement Learning for Intelligent Transportation Systems: A Survey	Haydari	2022	IEEE Transactions on Intelligent Transportation Systems	review (E3; used for chaining)
G20	Traffic Signal Control via Reinforcement Learning: A Review on Applications and Innovations	Michailidis	2025	Infrastructures	review (E3; used for chaining)

Appendix B. Extraction codebook

B.1 Descriptive fields (one row per paper)

Bibliographic

Field	What is recorded	Allowed values
paper_id	Stable paper ID (P001…)	free text
authors	Authors	free text
year	Publication year	whole number
title	Title	free text
venue	Journal or conference	free text
venue_type	Venue type	journal, conference
doi	DOI	free text
country_first_author	Country of first author's affiliation (pre-filled guess, verify)	free text
study_location	City/country of the studied network, or 'synthetic'	free text

Application

Field	What is recorded	Allowed values
application_context	Spatial setting of the controlled intersections	isolated_intersection, arterial_corridor, grid_network, real_urban_network, other
n_intersections	Largest number of RL-controlled intersections evaluated	whole number
traffic_modes	Road users explicitly modelled/prioritised	private_vehicles, pedestrians, cyclists, public_transport, emergency_vehicles, cav_connected_vehicles, freight, mixed
objective_focus	Objectives explicitly optimised	efficiency, safety, emissions_energy, equity_fairness, priority, robustness

RL formulation and technology

Field	What is recorded	Allowed values
rl_family	RL algorithm family of the proposed method	tabular, value_based_deep, policy_gradient, actor_critic, model_based, hierarchical, meta_rl, offline_rl, llm_rl_hybrid, other
algorithm	Named algorithm (e.g. DQN, D3QN, PPO, A2C, MAPPO)	free text
agent_architecture	Agent architecture	single_agent, multi_agent_independent, multi_agent_cooperative, centralised_multi_intersection
coordination	Coordination mechanism between intersections	none, neighbour_observation, explicit_communication, graph_attention, pressure_based, hierarchical, other

Field	What is recorded	Allowed values
state_repr	State features	queue_length, waiting_time, vehicle_count, occupancy, speed, position_matrix_image, pressure, current_phase, phase_duration, cv_trajectory, raw_camera, other
action_type	Action definition	phase_selection, phase_switch_keep, phase_duration, cycle_split, offset, other
reward_type	Reward components	queue_length, waiting_time, delay, pressure, throughput, travel_time, speed, emissions, safety, composite, other
function_approximator	Policy/value function approximator	table, mlp, cnn, rnn_lstm, gnn, attention_transformer, llm, other
simulator	Simulation environment(s)	sumo, cityflow, vissim, aimsun, paramics, cbengine, custom, other, none
sensing_source	Sensing assumed or used for the state	loop_detector, camera, radar_lidar, connected_vehicle, simulator_ground_truth, other

Validation

Field	What is recorded	Allowed values
network_source	Road network geometry	synthetic, real
demand_source	Traffic demand used in evaluation	synthetic, real_dataset, calibrated, field_live
dataset_ids	Benchmark datasets/scenarios, ';'-separated canonical names (e.g. jinan_3x4; hangzhou_4x4; ny_28x7; monaco; ingolstadt7; cologne8)	free text
baseline_classes	Classes of baselines compared	fixed_time, actuated, max_pressure, sotl, scats_scoot, other_adaptive, rl, other
vrl	Validation Realism Level 0-5 (highest reached)	0–5
transfer_test	Evaluated on unseen scenarios / other simulator	Y, N
robustness_test	Detector noise/failure, demand surge or incident tests	Y, N
safety_constraints	Min green, clearance, action masking or other constraints enforced	Y, N
n_seeds	Number of independent runs/seeds ('NR' if not reported)	free text
code_available	Code publicly available	Y, N

Conclusions

Field	What is recorded	Allowed values
key_findings	Main findings (1-2 sentences)	free text
advantages	Advantages claimed	free text
limitations_authors	Limitations stated by the authors	free text

Reviewer interpretation

Field	What is recorded	Allowed values
barriers	Deployment barriers discussed or evident	sensing_data, sim_to_real_gap, safety_compliance, robustness, computation, communication, interpretability, scalability, multimodality, training_cost_sample_efficiency, regulation_institutional, none_reported
mitigations	Mitigations proposed for the barriers	free text
reviewer_strength	Main methodological strength (reviewer)	free text
reviewer_weakness	Main weakness (reviewer)	free text
research_gap_note	Gap this study reveals or leaves open	free text

Dependence control

Field	What is recorded	Allowed values
study_family_id	Same experiments/project/dataset family -> same ID (e.g. F01). Default = paper_id	free text
campaign_id	Field campaign or shared experiment ID, if any	free text

Record keeping

Field	What is recorded	Allowed values
extracted_by	Extractor	R1, R2
verified_by	Verifier (20 % sample)	R1, R2

B.2 Numerical fields (one row per reported value)

Field	What is recorded	Allowed values
entry_id	Q0001…	free text
paper_id	Paper ID	free text
scenario_id	Dataset/scenario canonical name for this number	free text
metric	Outcome metric	travel_time, waiting_time, delay, queue_length, stops, fuel, co2, emissions_other, safety_conflicts, throughput, speed, other
rl_method_name	Name of the proposed RL method	free text
rl_value	Value for the proposed RL method	number
baseline_name	Baseline name as in the paper	free text
baseline_class	Baseline class	fixed_time, actuated, max_pressure, sotl, scats_scoot, other_adaptive, rl, other

Field	What is recorded	Allowed values
baseline_value	Value for the baseline	number
reported_improvement_pct	Use only if the paper gives the % improvement but not both raw values	number
unit	Unit (s, veh, veh/h, m, g…)	free text
rl_sd	SD/dispersion of RL value, if reported	number
baseline_sd	SD of baseline value, if reported	number
n_runs	Number of runs behind the values	whole number
vrl	Validation level of this experiment (defaults to paper VRL)	0–5
context	Demand level, peak/off-peak, test condition…	free text
source_location	Page/table/figure, e.g. 'Table 4, p. 9'	free text
extracted_by	Extractor	R1, R2
verified_by	Verifier (100 % of entries)	R1, R2